EE3032 Introduction to VLSI Design – Final

Fall 2013

Note: Total number of points is 150, but the number of points you can get is smaller than 100. The printed note and all point labels are preserved without rescaling.

1. (40 points) Answer the following questions.

(a) List two approaches which can be used to alleviate the occurrence probability of latchup. (4 points)

(b) Explain the following definitions: (1) physical design, (2) instantiate a cell, and (3) instance. (6 points)

(c) List two requirements which must be met when you draw the layout of a cell in the cell library. (4 points)

(d) What is soft error? (2 points)

(e) Why is the soft error rate of a circuit increased with increasing operation frequency and scaling down the technology process? Explain the cause in terms of logical masking effect, electrical masking effect, and latch window effect. (6 points)

(f) List two methods which can be used to reduce the power bounce and ground bounce occurring in the IO pads. (4 points)

(g) List two major problems of dynamic logic. (4 points)

(h) List two limitations of CMOS domino logic. (4 points)

(i) What are core-limited design and pad-limited design? (4 points)

(j) A pseudo-NMOS logic is called a ratioed logic. Why? (2 points)

2. (10 points) Sketch the Boolean function $f = \overline{AB(C+D)}$ using each of the following logic design techniques: (1) Dynamic logic (5 points) and (2) Fully differential logic (5 points).

3. (15 points) Figure 1 shows a domino logic gate. Assume that 0.18 μm technology is used to design the gate and $\mu_n = 3\mu_p$. Also, a $2\,\mu$m/$0.18\,\mu$m PMOS/NMOS has input capacitance C (i.e., $C_g = C$). Complete the following questions.

(a) Assume that the size of P1 is $4\,\mu$m/$0.18\,\mu$m and all channel lengths are $0.18\,\mu$m. Determine the sizes of N1–N6 such that the rise time and fall time of node X are equal, i.e., $t_r = t_f$. (5 points)

(b) Your boss asks you to design the dynamic gate such that the switching power of CLK is $2fCV_{DD}^2$, where f and V_{DD} denote the operation frequency and supply voltage. Determine the sizes of N1–N6 and P1 such that the rise time and fall time of node X are equal. (10 points)

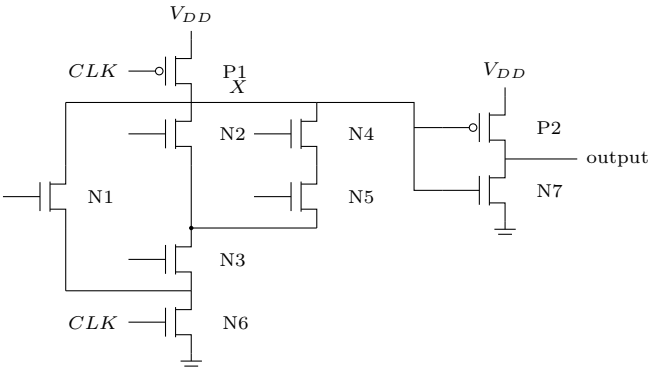


Figure 1: A dynamic logic circuit. Data gates are unlabeled as in the source.

4. (15 points) Figure 2 shows a combinational circuit designed with NP-domino logic. All NMOS have the same β_n, all PMOS have the same β_p, and $\beta_n = \beta_p$. Also $\mu_n = 2\mu_p$, the PMOS/NMOS gate capacitances are $2C/C$, respectively, and each turn-on equivalent resistance is R.

(a) The rise/fall time is the product of the output capacitance and equivalent resistance of the output charging/discharging path, respectively. The gate delay is $t_d = (t_r + t_f)/2$, and transistor drain capacitances can be neglected. Calculate the contamination delay t_{cd} and propagation delay t_{pd} in terms of R and C. (10 points)

(b) Assume $V_{DD} = 1.8$ V and $|V_{tp}| = 0.5$ V. The capacitances of x, y, z are $0.4, 0.8, 1.0$ pF, respectively. Can the circuit operate correctly? Explain. (5 points)

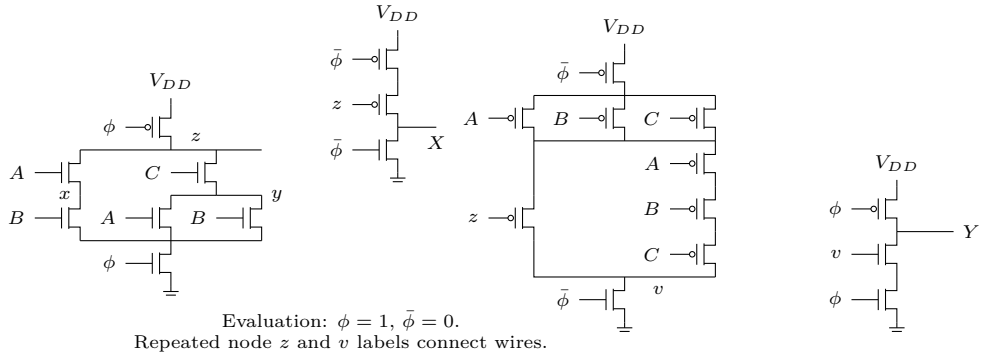


Figure 2: A circuit designed with NP-domino logic.

5. (25 points) Figure 3 shows a sequential circuit. The register clock-to-Q propagation delay, clock-to-Q contamination delay, setup time and hold time are $0.3, 0.1, 0.2, 0.4$ ns, respectively. The NOR, XOR, XNOR and inverter delays are $0.2, 0.5, 0.4, 0.2$ ns, respectively.

(a) If clock skew is 0 ns, can the circuit operate correctly? If not, modify the circuit so that it can operate correctly without sacrificing operation frequency. (5 points)

(b) If clock skew is 0 ns, determine the maximum operation frequency. (5 points)

(c) If clock skew is 0.5 ns, determine the maximum operation frequency. (5 points)

(d) If you can insert three additional registers to boost the operation frequency, where should they be inserted so that maximum improvement is achieved? Draw the modified circuit and calculate its maximum delay. (10 points)

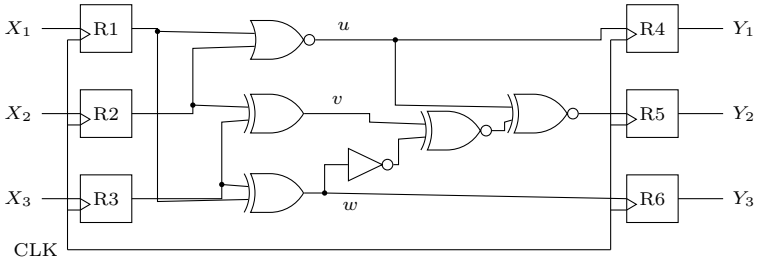


Figure 3: A sequential circuit.

6. (20 points) Figure 4 shows two symbolic layouts of two circuits. Draw the transistor-level schematic diagrams of the two circuits.

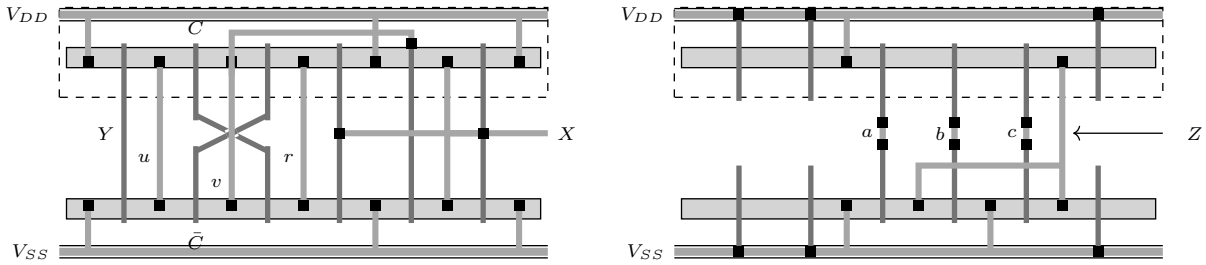


Figure 4: Symbolic layouts of two circuits: (a) left, (b) right.

7. (10 points) Figure 5 shows a Schmitt circuit, where $V_{tn} = |V_{tp}| = 0.6$ V. If an input signal as shown in the left figure is applied, the circuit outputs the signal shown in the right figure. Derive the ratios β_{N1}/β_{N3} and β_{P1}/β_{P3}.

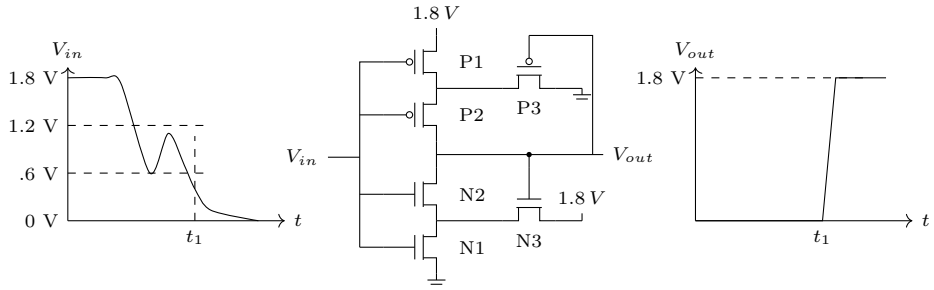
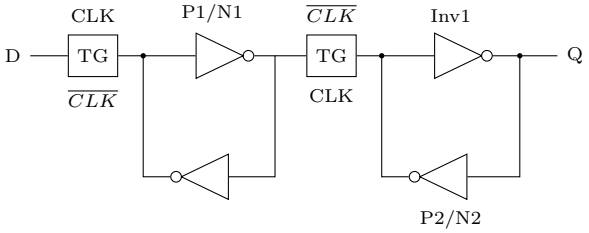


Figure 5: A CMOS Schmitt circuit.

8. (15 points) Figure 6 shows a register realized by two low-cost latches. Complete the following questions.

(a) What type of register is it? Explain. (5 points)

(b) Assume that the switching threshold of inverter Inv1 is $V_{DD}/2$, and the turn-on equivalent resistances of the transmission gate, N1, N2, P1 and P2 are $R_{TG}, R_{N1}, R_{N2}, R_{P1}, R_{P2}$, respectively. List two inequalities for determining the sizes of P2 and N2 such that the register can operate correctly. (10 points)



Upper/lower TG controls are PMOS/NMOS gates.
Both feedback inverters remain connected.

Figure 6: A register consisting of two low-cost latches.

Reconstruction notes: Three dated source photographs are retained in original-exam.pdf. All reconstructed figures are editable vectors. Identical 2012 figures were consulted only for low-resolution circuit/contact details; 2013 wording, numbers and points govern.